

add time
dimension

study

+ play

SYSTEMS SIMULATION

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

validation

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

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FRANCISCO JOSÉ DE CALDAS

Outline

1 Basic Concepts

2 Cellular Automata

3 Digital Twins



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- 3 Digital Twins



What is a simulation?

- **Simulations** are a type of real-world representation, but rarely exactly like the real world.
- Sometimes you need to **test** or **experiment** with expensive use cases. Simulations let you **play** with different inputs, configurations, hyperparameter optimizations.
- Also, there are dangerous or hard-to-reach **scenarios** where **simulations** become the best option.
- You should define the **level of detail** well enough to represent the expected behavior, without failing due to either high complexity or lazy simplicity.

high complexity



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chemical processes construction



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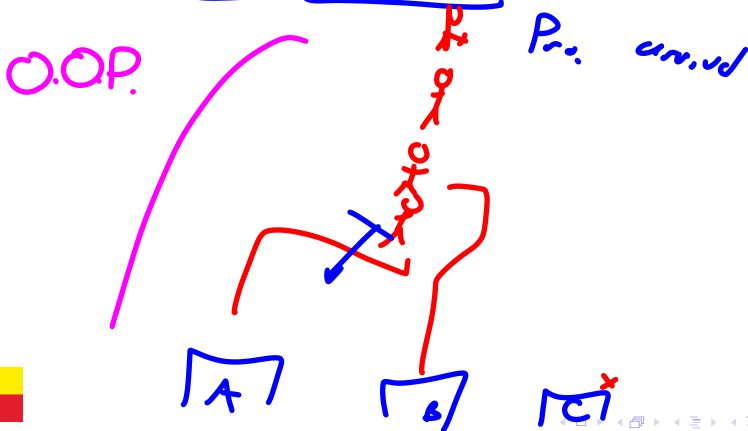
Events and Stochastic Processes

- It is typical to play with **event probabilities**, creating **stochastic behaviors**.
- One way to **simulate many systems** is to use **event-based models**.
Embrace the chaos and Murphy's Law.

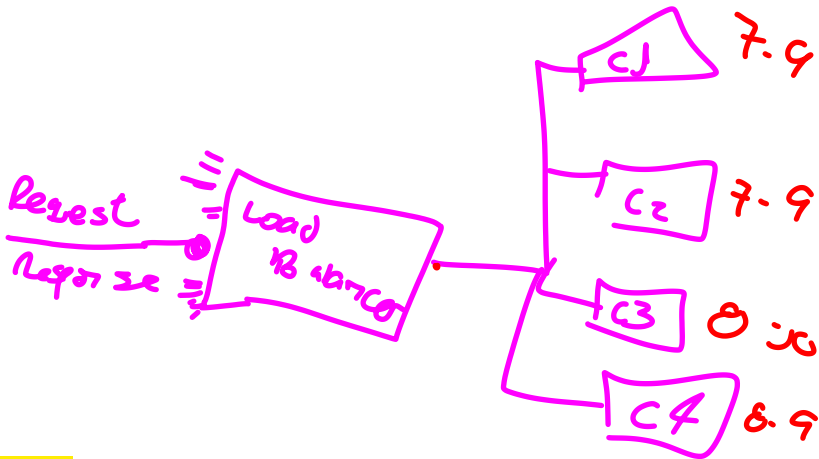


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Case Study: Load Balancer in a Cloud Architecture



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Cellular Automata

state changes

- Cellular Automata are a discrete model defined by a grid of cells, each with a state.
- The state of a cell is updated based on the states of its neighbors.

cells

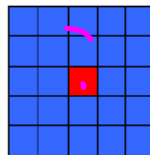
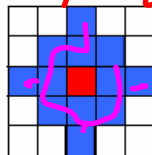
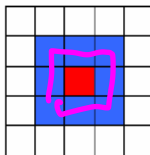
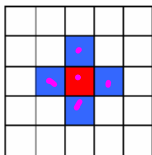
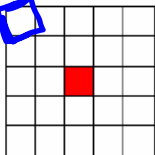


Figure: Cellular Automata Typical Neighborhoods



flow



Game of Life

- **Game of Life** is a **cellular automaton** devised by the British mathematician **John Horton Conway** in **1970**.
- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.
- Rules:

chaos

$t = 0$

- Any live cell with fewer than two live neighbors dies, as if by underpopulation.
- Any live cell with two or three live neighbors lives on to the next generation.
- Any live cell with more than three live neighbors dies, as if by overpopulation.
- Any dead cell with exactly three live neighbors becomes a live cell, as if by reproduction.



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- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.
- Rules: *→ deterministic*
 - Any **live** cell with **fewer than two** live neighbors **dies**, as if by underpopulation. *0 - dead, 1 - live*
 - Any **live** cell with **two or three** live neighbors **lives** on to the next generation.
 - Any **live** cell with **more than three** live neighbors **dies**, as if by overpopulation.
 - Any **dead** cell with **exactly three** live neighbors becomes a **live** cell, as if by reproduction.

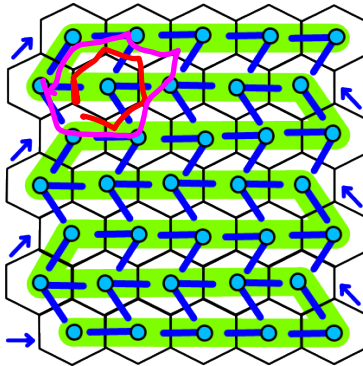


for $i \in \{ \dots \}$
 for $j \in \{ \dots \}$
 $sum = \sum_{i,j} M_{ij}$



HoneyComb Cellular Automata

- **HoneyComb Cellular Automata** is a different topology where a cell has six neighbors.
- This representation has different dispersion properties, which can be more interesting.



→ sparse matrix



SIR Model

→ SARS (2004)

→ discrete

- **SIR Model** is a compartmental model used to represent the transmission of a contagious disease.

- The **model** divides the population into three compartments: **S** for the number of susceptible individuals, **I** for the number of infected individuals, and **R** for the number of recovered individuals.

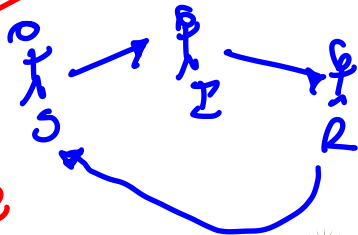
- The model is defined by the following differential equations where β is the transmission rate and γ is the recovery rate:

$$\frac{dS}{dt} = -\beta \cdot S \cdot I$$

$$\frac{dI}{dt} = \beta \cdot S \cdot I - \gamma \cdot I$$

$$\frac{dR}{dt} = \gamma \cdot I$$

Prob.



↓
~ Game Of Life

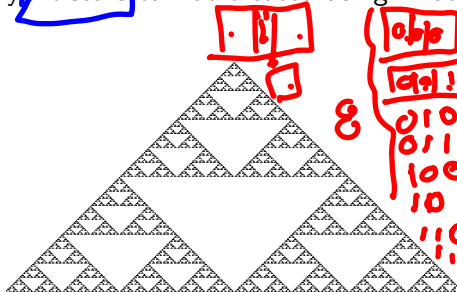


Chaotic Systems

- **Chaotic Systems** are a class of dynamical systems that exhibit sensitive dependence on initial conditions. *8 bits $\rightarrow$ 1 byte*
- This means that the future behavior of the system is highly dependent on the initial conditions.
- The **Lorenz System** is a well-known example of a chaotic system.
- Using **cellular automata** to simulate chaotic systems is a common practice. Many **fractals** can be created using what are called chaotic rules.

$00011000 = 24$
 $01080000 = 10$
 8 bits $\rightarrow$ 1 byte

4D $\sim$ 3.7D
 3D $\sim$ 2.1D
 2D $\sim$ 1.9D



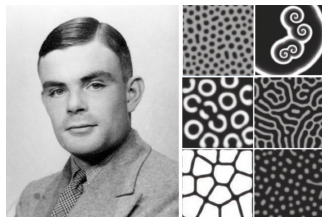
Sierpinski

0000 = 0
 0001 = 1
 0010 = 2
 0011 = 3
 0100 = 4
 0101 = 5
 0110 = 6
 0111 = 7
 1000 = 8
 1001 = 9
 1010 = 10
 1011 = 11
 1100 = 12
 1101 = 13
 1110 = 14
 1111 = 15



Turing Morphogenesis

- **Turing Morphogenesis** is a theory of **biological development** that explains how **patterns** form in **living organisms**.
- The theory is based on the idea that **chemical signals** can interact to create **patterns** in living organisms.
- The **reaction-diffusion** model is a common way to simulate **Turing morphogenesis**.
- The model is defined by a set of **reaction** and **diffusion** equations that describe how the **chemical signals** interact.



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What are Digital Twins?

- A **Digital Twin** is a *virtual model* that accurately reflects a physical object or **system**.
- **Digital twins** enable **real-time monitoring**, **simulation**, and **analysis** of the physical counterpart.
- They are widely used in **manufacturing**, **smart cities**, **healthcare**, and **energy systems**.
- By integrating sensor data, **digital twins** help in **predictive maintenance** and **process optimization**.



Applications of Digital Twins

- **Manufacturing:** Monitor equipment performance and **predict failures**.
- **Urban Planning:** Model city infrastructure to **optimize traffic** and resource use.
- **Healthcare:** **Simulate** patient-specific **treatments** and optimize procedures.
- **Energy Systems:** Analyze **grid performance** and integrate renewable energy sources.



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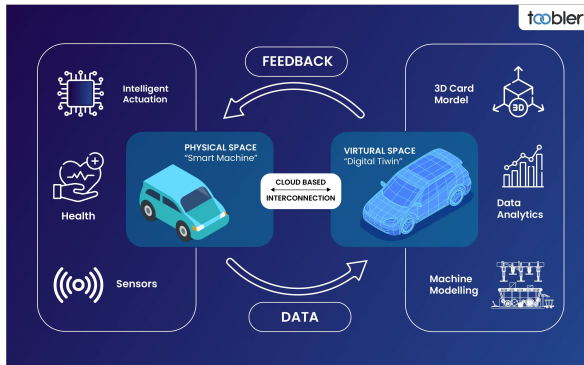
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Benefits of Digital Twins

- Enhanced **decision-making** through **real-time data analysis**.
- Improved efficiency and reduced operational costs.
- *Better understanding* of **system behavior** and **performance**.
- Increased collaboration between teams and stakeholders.



Challenges of Digital Twins

- Data **integration** and **management**.
- Ensuring data security and privacy.
- Maintaining the accuracy of the digital twin.
- Managing the complexity of the physical system.
- Ensuring interoperability between different systems and platforms.
- Scalability to accommodate large-scale systems.
- Cost of implementation and maintenance.
- User training and adoption.
- Regulatory compliance and standardization.
- Ethical considerations in data usage and analysis.
- Environmental impact of digital twin technologies.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

